星际介质天文学作业

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22.1 (a) 因为 $\lambda = 10\,\mu\text{m} \ll a = 0.1\,\mu\text{m}$，可以认为这里的尘埃可以用电偶极近似，由 (22.19) 式得：

$$C_{\text{abs}} = 18\pi \frac{\varepsilon_2}{(\varepsilon_1+2)^2+\varepsilon_2^2} \cdot \frac{V}{\lambda} = 18\pi \times \frac{2.575}{(0.996+2)^2+2.575^2} \times \frac{(4\pi/3)a^3}{\lambda} = 3.908\times10^{-3}\,\mu\text{m}^2$$

则有：

$$Q_{\text{abs}} = \frac{C_{\text{abs}}}{\pi a^2} = 0.124$$

且：

$$\frac{C_{\text{abs}}}{V} = \frac{C_{\text{abs}}}{(4\pi/3)a^3} = 0.933\,\mu\text{m}^{-1}$$

22.1 (b) 首先使用 (22.20) 式计算 C_{sca}，注意这里的 ε 是复数：

$$C_{\text{sca}} = 24\pi^3 \left|\frac{\varepsilon-1}{\varepsilon+2}\right|^2 \frac{V^2}{\lambda^4} = 24\pi^3 \times \left|\frac{-0.004+2.575i}{2.996+2.575i}\right|^2 \times \frac{\left[(4\pi/3)a^3\right]^2}{\lambda^4} = 5.547\times10^{-7}\,\mu\text{m}^2$$

则有：

$$Q_{\text{sca}} = \frac{C_{\text{sca}}}{\pi a^2} = 1.766\times10^{-5}$$

22.2 (a) 由 (22.12) 和 (22.14) 式，当尘埃的尺度远小于电磁波的波长时，吸收截面为：

$$C_{\text{abs},j} = \frac{4\pi\omega}{c}\,\text{Im}(\alpha_{jj}) = \frac{4\pi\omega}{c}\,\text{Im}\left[\frac{V}{4\pi} \cdot \frac{\varepsilon-1}{(\varepsilon-1)L_j+1}\right]$$

首先计算 $\text{Im}(\alpha_{jj})$，需要将 α_{jj} 的实部与虚部分离：

$$\begin{aligned}
\alpha_{jj} &= \frac{V}{4\pi} \cdot \frac{\varepsilon-1}{(\varepsilon-1)L_j+1} \\
&= \frac{V}{4\pi} \cdot \frac{(\varepsilon_1-1)+i\varepsilon_2}{1+(\varepsilon_1-1)L_j+i\varepsilon_2 L_j} \\
&= \frac{V}{4\pi} \cdot \frac{[(\varepsilon_1-1)+i\varepsilon_2]\cdot[1+(\varepsilon_1-1)L_j-i\varepsilon_2 L_j]}{[1+(\varepsilon_1-1)L_j]^2+(\varepsilon_2 L_j)^2} \\
&= \frac{V}{4\pi} \cdot \frac{(\varepsilon_1-1)\cdot[1+(\varepsilon_1-1)L_j]+\varepsilon_2^2 L_j+i\varepsilon_2}{[1+(\varepsilon_1-1)L_j]^2+(\varepsilon_2 L_j)^2}
\end{aligned}$$

则其虚部为：

$$\text{Im}(\alpha_{jj}) = \frac{V}{4\pi} \cdot \frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_j]^2+(\varepsilon_2 L_j)^2}$$

于是有：

$$C_{\text{abs},j}=\frac{4\pi\omega}{c}\text{Im}(\alpha_{jj})=\frac{\omega V}{c}\cdot\frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_j]^2+(\varepsilon_2L_j)^2}$$

又因为 $\omega=2\pi\nu$、$\lambda\nu=c$，则 $\dfrac{\omega}{c}=\dfrac{2\pi}{\lambda}$，故：

$$C_{\text{abs},j}=\frac{2\pi V}{\lambda}\cdot\frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_j]^2+(\varepsilon_2L_j)^2}$$

22.2 (b) 由题意：

$$\begin{aligned}
C_{\text{abs},a}-C_{\text{abs},b}&=\frac{2\pi V}{\lambda}\times\left\{\frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_a]^2+(\varepsilon_2L_a)^2}-\frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_b]^2+(\varepsilon_2L_b)^2}\right\}\\
&=\frac{2\pi V\varepsilon_2}{\lambda}\times\frac{\{[1+(\varepsilon_1-1)L_b]^2+(\varepsilon_2L_b)^2\}-\{[1+(\varepsilon_1-1)L_a]^2+(\varepsilon_2L_a)^2\}}{\{[1+(\varepsilon_1-1)L_a]^2+(\varepsilon_2L_a)^2\}\cdot\{[1+(\varepsilon_1-1)L_b]^2+(\varepsilon_2L_b)^2\}}\\
&=\frac{2\pi V\varepsilon_2}{\lambda}\times\frac{\{[1+(\varepsilon_1-1)L_b]^2-[1+(\varepsilon_1-1)L_a]^2\}+\{(\varepsilon_2L_b)^2-(\varepsilon_2L_a)^2\}}{\{[1+(\varepsilon_1-1)L_a]^2+(\varepsilon_2L_a)^2\}\cdot\{[1+(\varepsilon_1-1)L_b]^2+(\varepsilon_2L_b)^2\}}\\
&=\frac{2\pi V\varepsilon_2}{\lambda}\times\frac{[2+(\varepsilon_1-1)L_b+(\varepsilon_1-1)L_a]\cdot[(\varepsilon_1-1)(L_b-L_a)]+\varepsilon_2^2(L_b+L_a)\cdot(L_b-L_a)}{\{[1+(\varepsilon_1-1)L_a]^2+(\varepsilon_2L_a)^2\}\cdot\{[1+(\varepsilon_1-1)L_b]^2+(\varepsilon_2L_b)^2\}}\\
&=\frac{2\pi V\varepsilon_2}{\lambda}(L_b-L_a)\times\frac{[2+(\varepsilon_1-1)(L_b+L_a)]\cdot(\varepsilon_1-1)+\varepsilon_2^2(L_b+L_a)}{\{[1+(\varepsilon_1-1)L_a]^2+(\varepsilon_2L_a)^2\}\cdot\{[1+(\varepsilon_1-1)L_b]^2+(\varepsilon_2L_b)^2\}}\\
&=\frac{2\pi V\varepsilon_2}{\lambda}(L_b-L_a)\times\frac{(\varepsilon_1-1)^2(L_b+L_a)+2(\varepsilon_1-1)+\varepsilon_2^2(L_b+L_a)}{\{[1+(\varepsilon_1-1)L_a]^2+(\varepsilon_2L_a)^2\}\cdot\{[1+(\varepsilon_1-1)L_b]^2+(\varepsilon_2L_b)^2\}}
\end{aligned}$$

22.3 (a) 首先计算 e：

$$e=\sqrt{1-\left(\frac{b}{a}\right)^2}=\frac{\sqrt{5}}{2}$$

因为 $a<b$，由 (22.16) 式得：

$$L_a=\frac{1+e^2}{e^2}\left(1-\frac{1}{e}\arctan e\right)=0.446$$

则：

$$L_b=\frac{1}{2}(1-L_a)=0.277$$

22.3 (b) 由 (22.2) 题得：

$$C_{\text{abs},j}=\frac{2\pi V}{\lambda}\cdot\frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_j]^2+(\varepsilon_2L_j)^2}$$

分别代入 L_a、L_b 得：

$$\frac{C_{\text{abs},a}}{V}=\frac{2\pi}{\lambda}\cdot\frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_a]^2+(\varepsilon_2L_a)^2}=0.699\,\mu\text{m}^{-1}$$

$$\frac{C_{\text{abs},b}}{V}=\frac{2\pi}{\lambda}\cdot\frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_b]^2+(\varepsilon_2L_b)^2}=1.074\,\mu\text{m}^{-1}$$

则：

$$\frac{C_{\text{abs},b}-C_{\text{abs},a}}{V}=1.074\,\mu\text{m}^{-1}-0.699\,\mu\text{m}^{-1}=0.375\,\mu\text{m}^{-1}$$

22.3 (c) 如果这些椭球形尘埃是随机分布的，平均的吸收截面应当是按比例分配：

$$\frac{\langle C_{\rm abs}\rangle}{V} = \frac{1}{3}\times\frac{C_{{\rm abs},a}}{V} + \frac{2}{3}\times\frac{C_{{\rm abs},b}}{V} = 0.949\,\mu{\rm m}^{-1}$$

由 (22.1) 题得同类型的 $a = 0.1\,\mu$m 球形尘埃在 $\lambda = 10\,\mu$m 的吸收截面为 $\frac{C_{\rm abs}}{V} = 0.933\,\mu{\rm m}^{-1}$，则 $a = 0.1\,\mu$m、$b = 0.15\,\mu$m 的椭球形尘埃的吸收截面比同类型的 $a = 0.1\,\mu$m 球形尘埃要偏大约 1.7%。

22.4 由题意，波长为 $\lambda = 450\,\mu$m 时，介电常数为：

$$\varepsilon \approx 11.6 + 3.0\times\left(\frac{100\,\mu{\rm m}}{450\,\mu{\rm m}}\right)i = 11.6 + 0.667i$$

即：

$$\varepsilon_1 = 11.6,\quad \varepsilon_2 = 0.667$$

与 (22.3) 题相同，对于这种椭球形的尘埃粒子有：

$$L_a = 0.446,\quad L_b = 0.277$$

由 (22.2) 题得：

$$C_{{\rm abs},j} = \frac{2\pi V}{\lambda}\cdot\frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_j]^2+(\varepsilon_2 L_j)^2}$$

分别代入 L_a、L_b 得：

$$\frac{C_{{\rm abs},a}}{V} = \frac{2\pi}{\lambda}\cdot\frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_a]^2+(\varepsilon_2 L_a)^2} = 2.831\,\mu{\rm m}^{-1}$$

$$\frac{C_{{\rm abs},b}}{V} = \frac{2\pi}{\lambda}\cdot\frac{\varepsilon_2}{[1+(\varepsilon_1-1)L_b]^2+(\varepsilon_2 L_b)^2} = 5.993\,\mu{\rm m}^{-1}$$

则：

$$\frac{C_{{\rm abs},b}}{C_{{\rm abs},a}} = \frac{C_{{\rm abs},b}}{V} \div \frac{C_{{\rm abs},a}}{V} = 2.117$$

因为 $\hat{a}\perp\boldsymbol{B}_0$，则垂直于 $\boldsymbol{B}_0$ 的方向，即垂直于 $\hat{a}$ 的方向，热辐射的偏振度为：

$$\begin{aligned}
p &= \frac{I_{{\rm abs},b}-I_{{\rm abs},a}}{I_{{\rm abs},b}+I_{{\rm abs},a}}\\
&= \frac{C_{{\rm abs},b}-C_{{\rm abs},a}}{C_{{\rm abs},b}+C_{{\rm abs},a}}\\
&= \frac{(C_{{\rm abs},b}\div C_{{\rm abs},a})-1}{(C_{{\rm abs},b}\div C_{{\rm abs},a})+1}\\
&= \frac{2.117-1}{2.117+1}\\
&= 0.358
\end{aligned}$$